Probing and prompting LLM's

Report - Assignment 3 part 1

Name: Maartje Pfauder
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Introduction

The goal of this assignment is to investigate whether a Large Language Model (LLM) represents a story closer to its book version or its movie adaptation. Because LLMs are trained on large collections of textual data, it is interesting to examine how they internally store narrative knowledge when multiple versions of the same story exist. This question is particularly relevant for digital humanities research, where computational methods are increasingly used to analyse narratives and adaptations. This project therefore investigates whether the internal representations of an LLM align more closely with the book or the movie version of *Harry Potter and the Order of the Phoenix* using probing and prompting methods.

Research question and hypotheses

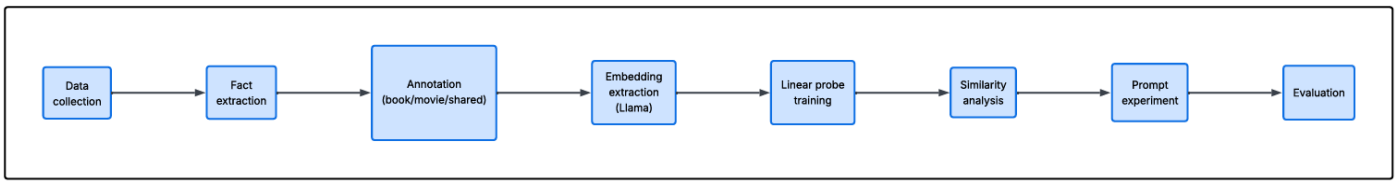
The original research question is too vague because it does not define what is meant by internal representation or how similarity between versions should be measured. The research question can therefore be reformulated as follows: *To what extent do the internal representations of an LLM more closely align with the book or the movie version of Harry Potter and the Order of the Phoenix, and does this vary across different types of narrative information such as plot facts, character roles, and tone?*
Based on this question, three hypotheses guide the study. First, the model is expected to be closer to the book version overall because LLMs are primarily trained on textual material. Second, representations of tone and emotional interpretation may show more blending between book and movie versions because these aspects are more interpretive. Third, later model layers are expected to better distinguish book-specific and movie-specific information because transformer models construct increasingly contextual representations across layers.

Research design

This study focuses on *Harry Potter and the Order of the Phoenix* because the adaptation differences between the book and the film are particularly pronounced. The novel is the longest in the series, while the movie is relatively short, resulting in major omissions and simplifications. Important differences include the removal of the St. Mungo's Hospital subplot, the absence of Ron's Quidditch storyline, and the reduced role of Kreacher. These differences make the story particularly suitable for examining whether an LLM encodes knowledge closer to one version than the other.
The dataset would consist of approximately 200 narrative facts extracted from the book, the movie, and adaptation comparison sources. To ensure reliability, a student annotator would label each fact according to whether it belongs to the book, the movie, or both versions, and would also categorize the type of information as plot, character, or tone. Ambiguous cases would be reviewed to ensure consistency. The legal department would be consulted to ensure copyright compliance, meaning only short factual summaries rather than copyrighted passages would be stored.

Methodology

Internal representation is defined in this study as the vector embeddings produced by the hidden layers of a transformer model when processing narrative facts. These embeddings capture semantic relationships and can therefore be analysed to determine whether the model internally distinguishes between book and movie knowledge. More specifically, internal representations are operationalized as hidden state vectors extracted from transformer layers, which encode contextual semantic information derived from the model's attention mechanisms. The study would use the open-source Llama 3.1 8B Instruct model through Ollama. The Llama 3.1 8B Instruct model was selected because it is a modern open-weight transformer model that allows access to internal layer representations, which is necessary for probing analysis.
The main experiment would involve converting each narrative fact into embeddings and extracting representations from an early, middle, and late layer of the model. All facts would be formatted as short declarative sentences to ensure consistency in input structure. A simple linear classifier would then be trained to predict whether a fact belongs to the book, the movie, or both. Linear probing is commonly used because if a simple classifier can distinguish categories, this suggests that the relevant information is already encoded in the representation space rather than requiring complex transformation. If classification is successful, this indicates that book and movie distinctions are linearly separable in the model's embedding space. In addition, cosine similarity would be used to measure whether representations cluster closer to book or movie facts, allowing similarity to be quantified.
Performance would be evaluated through classification accuracy, similarity comparisons, and differences between layers. Together these measures allow both quantitative evaluation and interpretive analysis of the model's internal structure.If later layers show clearer clustering or higher classification performance, this would support the hypothesis that deeper transformer layers encode more abstract narrative distinctions. The overall research procedure can be summarized in the following workflow:



Mini prompt experiment

To complement the probing design, a small prompt experiment was conducted using five example questions for which I know the book has a different answer from the movie. The five questions about The order of the phoenix were:

- Who told Harry about the Room of Requirement?
- What happens at St. Mungo's Hospital?
- Did Ron become a Quidditch hero?
- How is Harry's emotional struggle portrayed?
- What are important differences between the book and the movie?

Prompt type	**Expected answer**	**Model behaviour**	**Alignment**
Room of requirement	Dobby/Neville	Incorrect	Neither
St. Mungo's	Book subplot	Hallucinated	Neither
Ron Quidditch	Book hero	Incorrect	Neither
Emotional struggle	Psychological themes	Partially correct	Mixed
Differences	Adaptation changes	Partially correct	Mixed

Alignment refers here to whether the model response corresponds primarily to the book version, the movie version, both, or neither. The results show that the model frequently produced incorrect or blended answers. For example, when asked who told Harry about the Room of Requirement, the model produced an incorrect answer that did not correspond to either version. Similarly, the St. Mungo's question resulted in a fabricated explanation. When asked about Ron's Quidditch career, the model instead described unrelated heroic behaviour.
The model performed somewhat better on general interpretation questions. When asked about Harry's emotional struggles, it identified themes such as grief and isolation, although it also included invented details. When asked about differences between the book and the movie, the model correctly noted missing subplots but also included incorrect examples from other books.

Preliminary results

These preliminary results suggest that the model does not consistently align with either the book or movie version, but instead appears to contain blended and sometimes inconsistent narrative knowledge. The model appeared more reliable when discussing general themes than when recalling specific adaptation differences, suggesting that thematic understanding may be better represented than precise factual knowledge. The frequent hallucinations further indicate that prompting alone may not be sufficient to understand how narrative knowledge is structured internally.
These observations support the need for representation probing. While prompt responses were often inconsistent, it remains possible that internal representations are more clearly structured than the generated outputs suggest. The full study would therefore investigate whether internal layers contain clearer distinctions between book and movie knowledge than surface-level responses reveal. This also suggests that generated responses may not always directly reflect internal knowledge structure, highlighting the importance of analysing representations rather than relying only on output behaviour.

Limitations and contributions

Several limitations should be considered. It is unknown whether the model directly encountered the book or movie in its training data, and the model may reflect the most commonly discussed version rather than the most detailed one. Furthermore, annotation decisions may involve some subjectivity, and prompt responses may vary depending on wording. Additional methodological limitations should also be considered. First, the relatively small dataset of approximately 200 facts may limit statistical power. Second, because only one annotator is used, inter-annotator agreement cannot be measured. Third, model outputs may vary depending on prompt wording despite attempts at standardization. Fourth, transformer models are probabilistic systems, meaning some variability may remain even under controlled settings. Despite these limitations, the study demonstrates how probing and prompting can be combined to investigate how LLMs represent narratives across media adaptations.

Conclusion

The preliminary experiment suggests that the LLM does not clearly prefer either the book or movie version but instead appears to contain a blended representation of the story. This supports the need for systematic probing to determine whether internal representations show clearer alignment than prompt responses. More broadly, this project demonstrates how computational methods can be used in digital humanities research to study narrative adaptation and knowledge representation in language models.